

HSE 712 Foundations of Biostatistics

Correlations

Contents

- Pearson's correlation
- Spearman's correlation

Let's revisit covariance

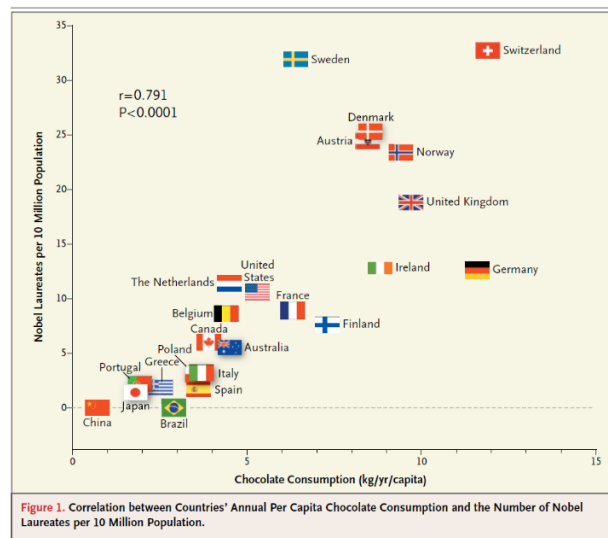
- Covariance is the average of the product deviations between two continuous random variables.

- Population value: $Cov(X, Y) = E[(X - \mu_x)(Y - \mu_y)]$

- Sample estimate: $Cov(\bar{X}, \bar{Y}) = s_{XY} = \frac{\sum_{i=1}^n (X_i - \bar{X})(Y_i - \bar{Y})}{n-1}$

- What are the units of measurement for a covariance?
 - Units for X * Units for Y
 - For example, a scatterplot of the number of Nobel laureates per 10 million population per country and per capita chocolate consumption (kg per year per capita) is presented below. (See [Messerli 2014 N Engl J Med 367;16:1562-1564](#))
 - The units for the covariance between those two data points would be

Nobel laureates per 10 million population per country x per capita chocolate consumption (kg per year per capita).



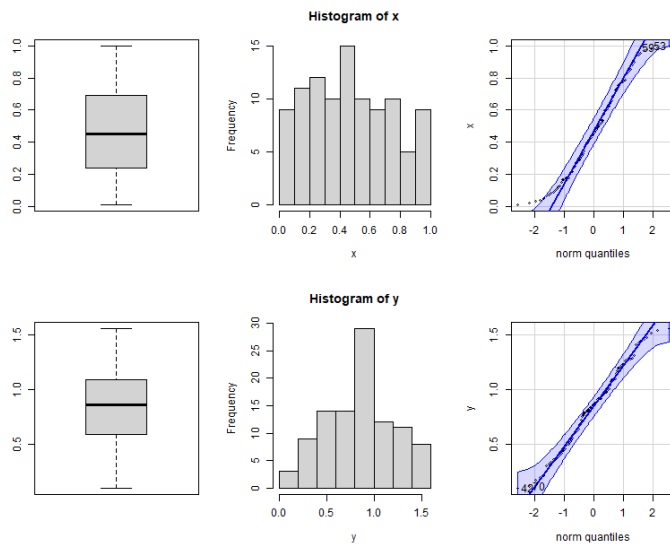
Because covariances are dependent on the units of measurement, it's hard to grasp what a "large" or "small" covariance is.

Example with r data

```
library(car)
x <- runif(100, 0, 1)
y <- 0.8*x + runif(100, 0, 1)
```

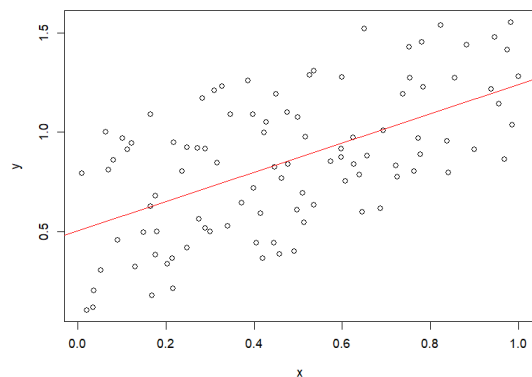
```
par(mfrow=c(2,3))
boxplot(x)
hist(x)
qqPlot(x)
```

```
boxplot(y)
hist(y)
qqPlot(y)
```



Both variables appear samples from a normal distribution.

```
par(mfrow=c(1,1))
plot(y~x)
abline(lm(y~x), col="red")
```

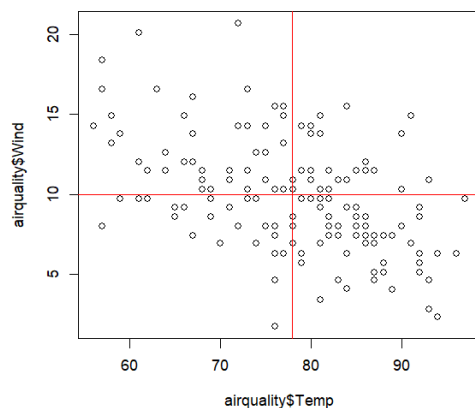


Looks like a linear bivariate association
cov(x,y)
0.0567 # X x Y units.

- If the random variables X and Y are independent, then the covariance between them is 0.
- If large values of X and Y tend to occur among the same subjects (as well as small values of X and Y), then the covariance is **positive**.
- If large values of X and small values of Y (or conversely, small values of X and large values of Y) tend to occur among the same subjects, then the covariance is **negative**.

```
# Example with air quality data (dataset within R)
# The R data set airquality includes daily air quality measurements in New
# York, May to September 1973.
?airquality
# Let's look at the associations between wind (mph) and ozone (ppb) with
# temperature (F)
# Wind: Average wind speed in miles per hour at 0700 and 1000 hours at
# LaGuardia Airport
# Ozone: Mean ozone in parts per billion from 1300 to 1500 hours at Roosevelt
# Island
# Temp: Maximum daily temperature in degrees Fahrenheit at La Guardia
# Airport.

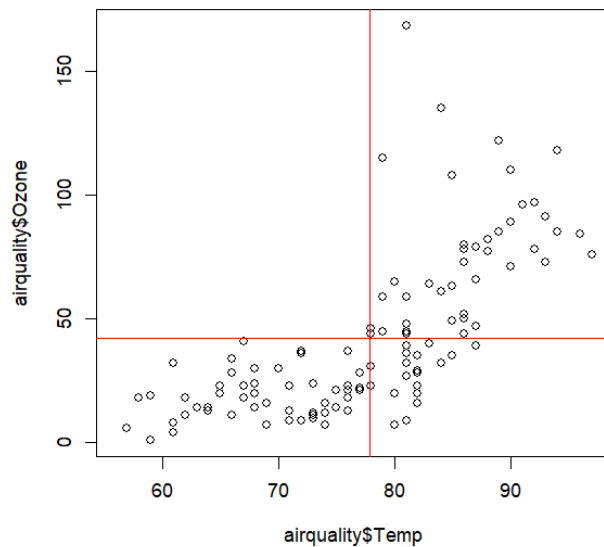
# wind vs. temp:
plot(airquality$wind ~ airquality$Temp)
abline(h=mean(airquality$wind, na.rm=T),col='red')
abline(v=mean(airquality$Temp, na.rm=T),col='red')
```



```
cov(airquality$wind , airquality$Temp)
# -15.27 mph*degF
# This is negative, so we know that as temperature increases, wind decreases
```

Ozone vs. temp:

```
plot(airquality$Ozone ~ airquality$Temp)
abline(h=mean(airquality$Ozone, na.rm=T),col='red')
abline(v=mean(airquality$Temp, na.rm=T),col='red')
```



```
cov(airquality$Ozone, airquality$Temp, use="complete.obs")
# 218.52 ppb*degF
# This is positive, so we know that as temperature increases, ozone measures
# increase
# NOTE - this appears slightly curvilinear.
```

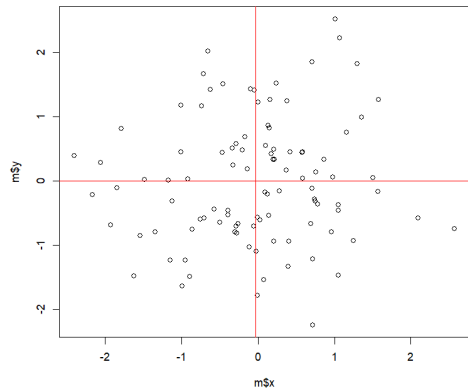
IN SUMMARY:

$\text{Cov}(\text{Wind}, \text{Temp}) = -15.3 \text{ mph} * \text{degF}$

$\text{Cov}(\text{Ozone}, \text{Temp}) = 218.5 \text{ ppb} * \text{degF}$

Can we tell which of these two associations is “stronger”?

```
# First, let's create an example where  $\text{cov}(X,Y) \approx 0$   
x <- rnorm(100, 0, 1)  
y <- rnorm(100, 0, 1)  
m <- as.data.frame(cbind(x,y))  
plot(m$y ~ m$x)  
abline(h=mean(m$y, na.rm=T),col='red')  
abline(v=mean(m$x, na.rm=T),col='red')
```



```
cov(m$y,m$x)  
# 0.08
```

Correlations

- Correlations are **standardized covariances**, and thus interpreting a correlation is much more intuitive.
- This class will review two common correlation coefficients
 - Pearson's correlation coefficient
 - Spearman's correlation coefficient

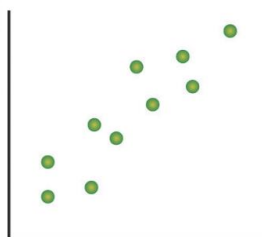
Pearson's correlation coefficient (Pearson's r)

- Conditions
 - Let X and Y be two continuous random variables.
 - Each random variable is sampled from a normal distribution.
 - X and Y are each i.i.d. and both are measured on the same sampled units.
- Assumptions
 - Pearson's correlation describes the strength and magnitude of the linear association between X and Y .



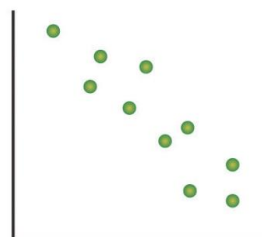
Pearson's correlation describes a bivariate distribution (between X and Y) that is assumed to be **linear**.

- There may be no bivariate association, and that is OK. However, this method is not OK with non-monotonic and non-linear associations.



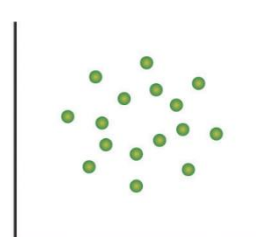
Positive linear
 $r = +.82$

OK to compute
Pearson's r



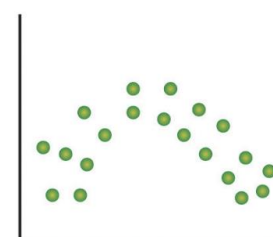
Negative linear
 $r = -.70$

OK to compute
Pearson's r



Independent
 $r = 0.00$

OK to compute
Pearson's r

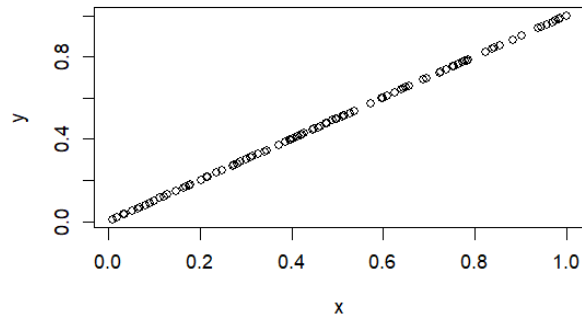


Curvilinear
 $r = 0.00$

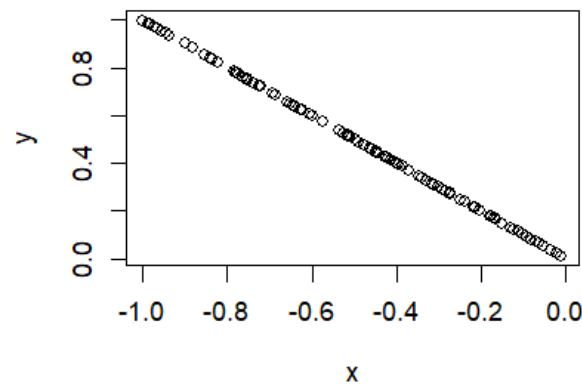
Not OK to compute
Pearson's r

ALWAYS PLOT THE DATA BEFORE COMPUTING CORRELATIONS!

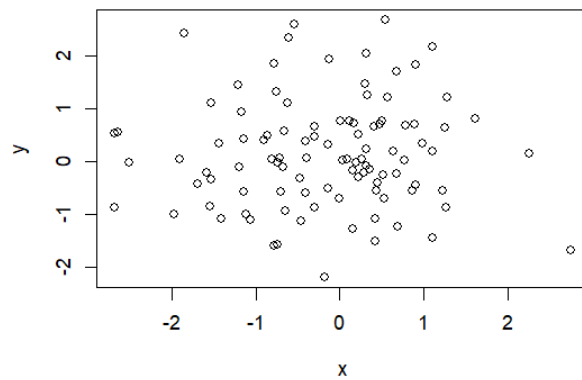
- A correlation coefficient ranges from -1.0 to 1.0.
 - 1.0 = A perfectly linear, positive association



- -1.0 = A perfectly linear, negative (or inverse) association



- 0.0 = No association, random scatter



- Pearson r is unitless because it is a **standardized measure of linear association**.

- Population value ($\rho = \text{rho}$):

$$\rho = \text{Corr}(X, Y) = \frac{\text{Cov}(X, Y)}{\sigma_X \sigma_Y}$$

Notice that the units in the numerator cancel out with the units in the denominator.

- Sample estimate:

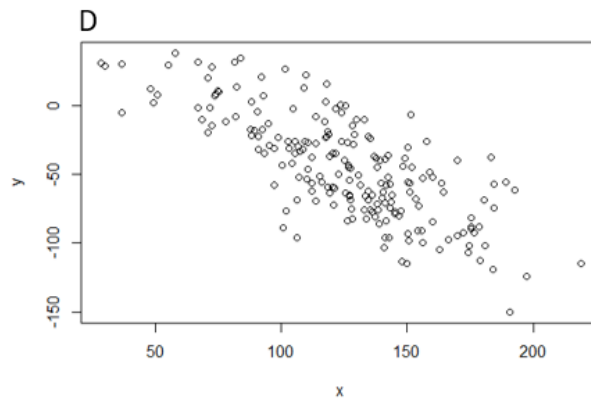
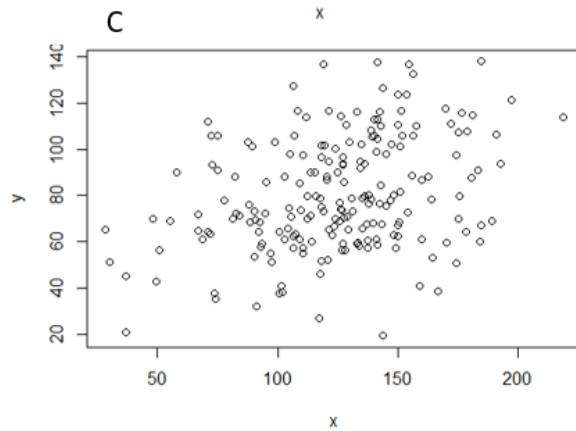
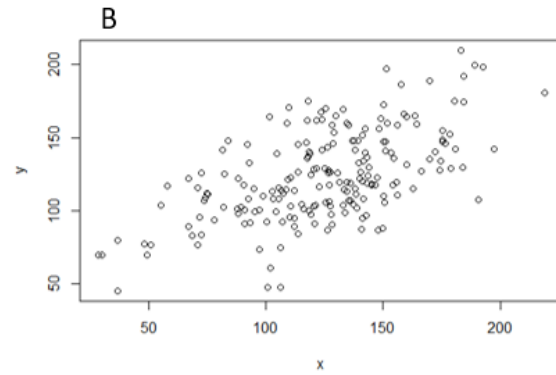
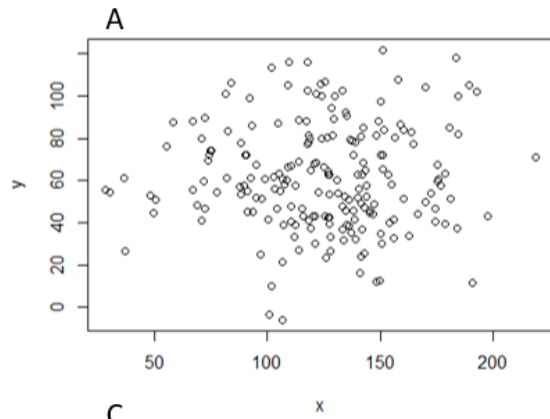
$$\hat{\rho} = r(X, Y) = \frac{\text{cov}(X, Y)}{s_X s_Y} = \frac{s_{XY}}{s_X s_Y}$$

Let's see this relationship in action:

```
set.seed(123456)
x <- rnorm(100, 0, 1)
y <- rnorm(100, 0, 1)
cov(x, y)
cov(x, y) / (sd(x) * sd(y))
cor(x, y)
```

```
> set.seed(123456)
> x <- rnorm(100, 0, 1)
> y <- rnorm(100, 0, 1)
> cov(x, y)
[1] -0.2098153
> cov(x, y) / (sd(x) * sd(y))
[1] -0.2146453
> cor(x, y)
[1] -0.2146453
```

Guess what the Pearson's Correlation would be for these plots (answers are on the next page):



Pearson's correlation	Figure
+ 0.23	C
+ 0.59	B
- 0.76	D
+ 0.02	A

Hypothesis Testing for Pearson's r

- Ho: There is no linear association between the two variables.
vs.
Ha: There is a linear association between the two variables.

Aka:

- Ho: $r = 0$
vs.
Ha: $r \neq 0$

A two-sided alternative hypothesis is the most common. One-side alternatives are possible, but rarely (if ever?) used in practice.

- Set alpha = 0.05 (let's be traditional).
- The test statistic for testing the null is a t test statistic.

$$t = \frac{r\sqrt{(n-2)}}{\sqrt{(1-r^2)}}$$

...where $t \sim t$ distribution with $n-2$ degrees of freedom and n = total sample size.

- How to interpret the final p-value?
 - If $p < 0.05$, reject the null hypothesis and conclude there is a linear association between X and Y. You must look at the data and the correlation coefficient to understand the direction and magnitude of that association.

- **A $(1 - \alpha)\%$ Confidence Interval** can also be created to demonstrate the precision in the estimated value of the true value of r .
 - This class won't emphasize the math for this interval, yet it is in Rosner for those who are interested (see Interval Estimation for Correlation Coefficients, 11.7).
 - The interpretation of the CI is the same as discussed before:
 - If the sampling scenario were repeated multiple times in the exact same way, then $(1 - \alpha)\%$ of the time this interval would include the true value of ρ in the population.
 - If the interval includes 0, then r is not statistically different from 0 at the α level.
 - If the interval does **not** include 0, then r **is** statistically different from 0 at the α level.

`cor.test()` in R provides a $(1 - \alpha)\%$ confidence interval, with the default being a 95% CI.
See the argument, `conf.level = 0.95`.

Extending the above example:

```
set.seed(123456)
x <- rnorm(100, 0, 1)
y <- rnorm(100, 0, 1)
cov(x,y)
cov(x,y) / (sd(x)*sd(y))
cor(x,y)
cor.test(x,y)
```

```
> set.seed(123456)
> x <- rnorm(100, 0, 1)
> y <- rnorm(100, 0, 1)
> cov(x,y)
[1] -0.2098153
> cov(x,y)/(sd(x)*sd(y))
[1] -0.2146453
> cor(x,y)
[1] -0.2146453
> cor.test(x,y)
```

```

      Pearson's product-moment correlation

data:  x and y
t = -2.1756, df = 98, p-value = 0.03199
alternative hypothesis: true correlation is not equal to 0
95 percent confidence interval:
-0.3944340 -0.0190295
sample estimates:
      cor
-0.2146453
```

The Pearson's correlation coefficient (r) for the bivariate linear association between x and y was -0.21, with a 95% CI -0.39 to -0.02. That interval does not include 0, and thus r is statistically different from 0 at the 0.05 (or 5%) level.

Another “full” example in R:

```
library(car) # For qq plot

# Create the data set
x <- rnorm(100, 0, 1)
y <- 0.8*x + runif(100, 0, 1)
# Assume x and y are both iid.

# Check assumption that each variable
# was sampled from a normal distribution

par(mfrow=c(2,3))
boxplot(x)
hist(x)
qqPlot(x)
# Looks good

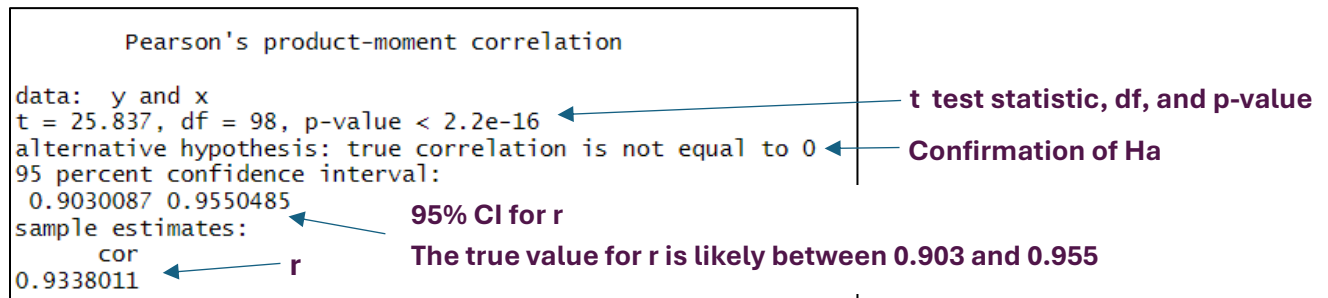
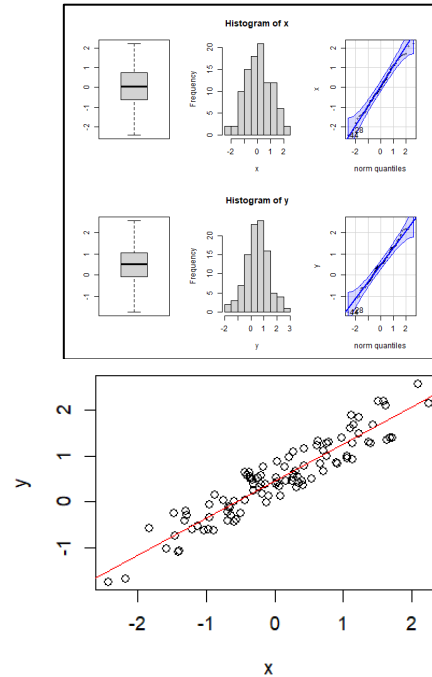
boxplot(y)
hist(y)
qqPlot(y)
# Looks good

# Check bivariate linear association
plot(y~x)
abline(lm(y~x), col="red")

# The plot supports a linear association
between X and Y. It specifically
supports a positive, linear association.

# Run statistical testing for Pearson's
correlation coefficient
# Ho:  $r = 0$  vs.  $H_a: r \neq 0$ .

cor.test(y, x, method = "pearson")
cor.test(y, x) # Pearson is the default - these two lines are the same
```



```
# There is a statistically significant, strong linear correlation
between X and Y (Pearson  $r = 0.93$ ;  $p < 0.001$ ).
```

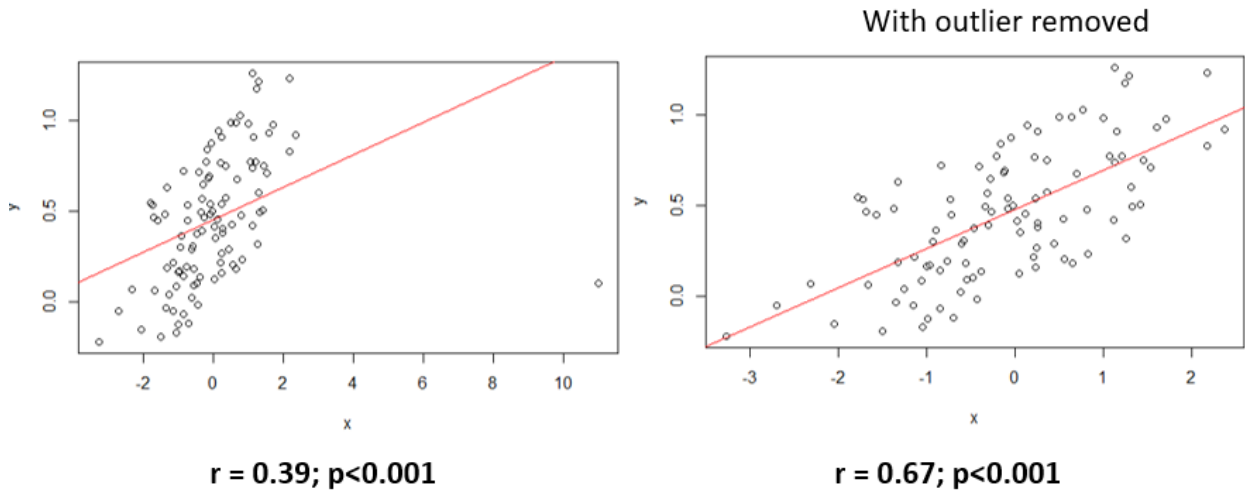
Magnitude of r

- In practice, we have recommendations for the “effect size” of r (Cohen 1988)
Cohen, J. (1988). *Statistical Power Analysis for the Behavioral Sciences*.

Strength of Association	Coefficient, r	
	Positive	Negative
Small	.1 to .3	-0.1 to -0.3
Medium	.3 to .5	-0.3 to -0.5
Large	.5 to 1.0	-0.5 to -1.0

Important Considerations about Pearson's correlation coefficient

- Pearson's correlation coefficient is sensitive to extreme values.



- The statistical significance of the hypothesis test, $H_0: r = 0$, is influenced by 1) the magnitude of the association and 2) the sample size.
 - Recall the test statistic for $H_0: r=0$ is a t test with $n-2$ df:

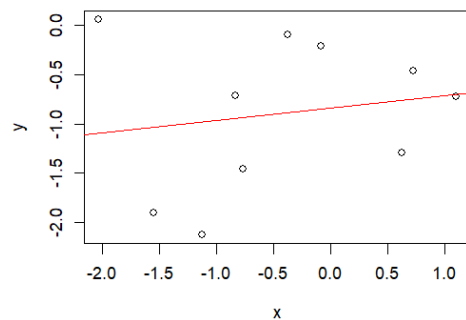
$$t = \frac{r\sqrt{(n-2)}}{\sqrt{(1-r^2)}}$$
 - For a given value of r , the t test statistic for the test $H_0: r = 0$ gets larger as the sample size (n) gets larger.
 - Thus, the p-value gets smaller.
 - Always consider the clinical or biological relevance of the effect size, regardless of the sample size.

GWAS studies may report small correlations between a genetic risk score and a phenotype (e.g., Alzheimer's disease, schizophrenia, obesity) that are statistically significant because GWAS studies typically include a very large sample size. For example, a Pearson's correlation of $r = 0.03$ would be statistically different from 0 at the 0.05 level with a sample size of 10,000. In that case, investigators need to consider if a correlation of 0.03 is meaningful.

Example in R

```
# For a given value of r, the p-value for the test  $H_0: r=0$  gets
# smaller as the sample size (n) gets larger.
# I set up the data to produce an  $r \approx 0.17$  in each case. Watch what
# happens to the statistical significance of testing  $H_0$ :
```

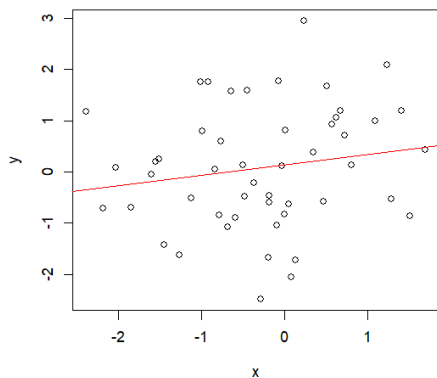
```
# n=10
set.seed(07202023)
x <- rnorm(10, 0, 1)
y <- 0.25*x + rnorm(10, 0, 1)
plot(y~x)
abline(lm(y~x), col="red")
cor.test(y, x)
```



```
Pearson's product-moment correlation

data: y and x
t = 0.48983, df = 8, p-value = 0.6374
alternative hypothesis: true correlation is not equal to 0
95 percent confidence interval:
 -0.5142359 0.7226274
sample estimates:
               cor
0.1706395
```

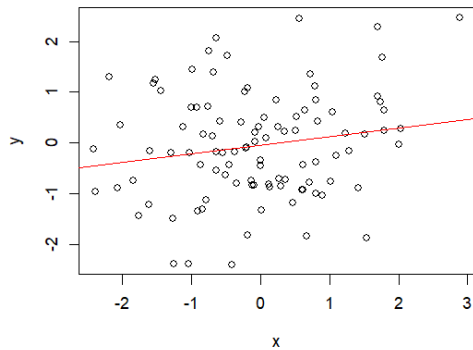
```
# n=50
set.seed(07202023)
x <- rnorm(50, 0, 1)
y <- 0.26*x + rnorm(50, 0, 1)
plot(y~x)
abline(lm(y~x), col="red")
cor.test(y, x)
```



```
Pearson's product-moment correlation

data: y and x
t = 1.1948, df = 48, p-value = 0.238
alternative hypothesis: true correlation is not equal to 0
95 percent confidence interval:
 -0.1137820 0.4280472
sample estimates:
               cor
0.1699481
```

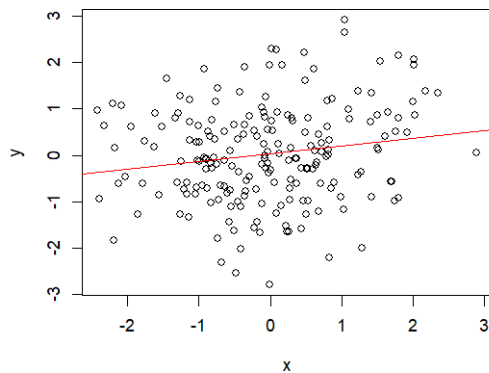
```
# n=100
set.seed(07202023)
x <- rnorm(100, 0, 1)
y <- 0.16*x + rnorm(100, 0, 1)
plot(y~x)
abline(lm(y~x), col="red")
cor.test(y, x)
```



Pearson's product-moment correlation

data: y and x
t = 1.7748, df = 98, p-value = 0.07903
alternative hypothesis: true correlation is not equal to 0
95 percent confidence interval:
-0.02066183 0.36039832
sample estimates:
cor
0.1764725

```
# n=200
set.seed(07202023)
x <- rnorm(200, 0, 1)
y <- 0.12*x + rnorm(200, 0, 1)
plot(y~x)
abline(lm(y~x), col="red")
cor.test(y, x)
```



Pearson's product-moment correlation

data: y and x
t = 2.4289, df = 198, p-value = 0.01604
alternative hypothesis: true correlation is not equal to 0
95 percent confidence interval:
0.03211434 0.30171810
sample estimates:
cor
0.1700975

Recap: Steps for computing Pearson's correlation

- Check the assumptions
 - Are the observations independent of each other?
 - Are the two random variables each from a normal distribution?
 - Is there a potential linear bivariate association between the two variables?
 - Are there any extreme values that may impact the value of r ?
- Compute r .
- Reject or do not reject the null.
- Interpret findings
 - Is the effect size clinically or biologically meaningful?

Spearman Rank Correlation Coefficient

- A non-parametric alternative to Pearson's correlation coefficient. The distribution for each random variable is not specified.
 - The distribution for the two variables is **not** assumed to be known and thus does not have to be normal.
- Let X and Y be two continuous random variables.
 - Many folks also use Spearman's r for ordinal data.
 - (FYI: Kendall's tau is a rank correlation also used for ordinal data)
- X and Y are each i.i.d. and both are measured on the same sampled units.
- Spearman's correlation describes the strength and magnitude of the monotonic association between X and Y.
 - It describes a bivariate distribution.
 - It assumes that bivariate distribution is monotonic, but it does **not** need to be linear.
 - There may be no bivariate association, and that is OK. We want to avoid non-monotonic associations.
- It is a standardized measure of a monotonic association.
 - It is unitless.
 - The correlation coefficient ranges from -1.0 to 1.0.
 - Sample estimate:

$$r_s = 1 - \frac{6 \sum_{i=1}^n d_i^2}{n^3 - n}$$

- Logic:
 - Rank each variable (e.g., X and Y)
 - d = difference in ranks between each variable within a sampled unit.
 - r_s is a metric of the sum of squared differences in ranked order between the two variables.
 - See extra slides for an example of this logic (optional). We will use R to compute correlations in this class; you do not need to compute these by hand.

Example

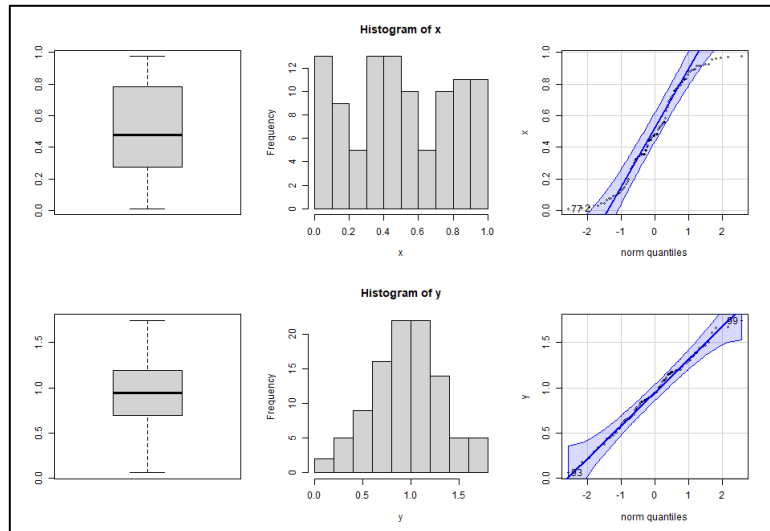
```
library(car)

# Generate the data
x <- runif(100, 0, 1)
y <- 0.8*x + runif(100, 0, 1)
```

```
# Plot each variable
par(mfrow=c(2,3))
boxplot(x)
hist(x)
qqPlot(x)
```

x does not look like it was sampled from a normal distribution (look at the tails in the QQ-plot).

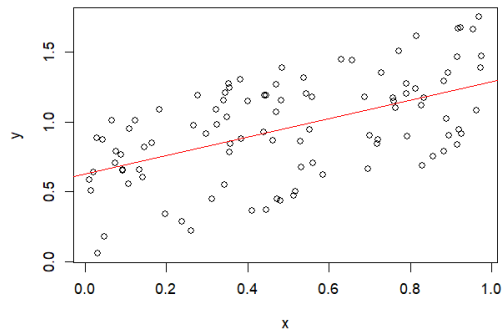
```
boxplot(y)
hist(y)
qqPlot(y)
```



y looks like it was sampled from a normal distribution.

Because x is likely not from a normal distribution, I choose to use Spearman's correlation.

```
# Bivariate association:
plot(y~x)
abline(lm(y~x), col="red")
```



Looks like a positive association. It also looks linear. However, I still want to use a Spearman's correlation because of the distribution of X.

```
# Computing the correlation and statistical testing:  
# I must include the argument for method = "spearman"
```

```
cor.test(y, x, method="spearman")
```

Spearman's rank correlation rho

```
data: y and x  
S = 79910, p-value = 4.629e-08  
alternative hypothesis: true rho is not equal to 0  
sample estimates:  
rho  
0.520492
```

```
# There is a statistically significant, moderate correlation between X  
and Y (Spearman r = 0.52; p<0.001).
```

Spurious Correlations

A spurious correlation is a correlation that appears in the data that is completely driven by random chance. Always be aware that spurious correlations could exist in your data!

The number of judges in Indiana

Detailed data title: BLS estimate of judges, magistrate judges, and magistrates in Indiana

Source: Bureau of Labor Statistics

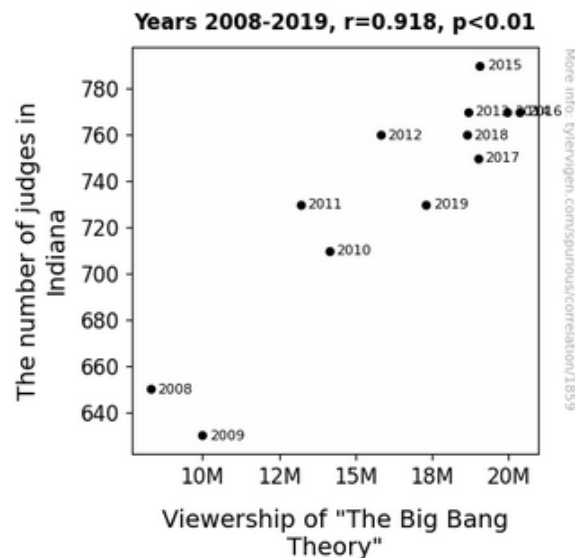
See what else correlates with The number of judges in Indiana

Viewership of "The Big Bang Theory"

Detailed data title: Average viewership of "The Big Bang Theory" by season

Source: Wikipedia

See what else correlates with Viewership of "The Big Bang Theory"



Correlation $r = 0.9176346$ (Pearson correlation coefficient)

Correlation is a measure of how much the variables move together. If it is 0.99, when one goes up the other goes up. If it is 0.02, the connection is very weak or non-existent. If it is -0.99, then when one goes up the other goes down. If it is 1.00, you probably messed up your correlation function.

Source: [TYLERVIGEN.COM](https://www.tylervigen.com)

https://www.tylervigen.com/spurious/correlation/1859_the-number-of-judges-in-indiana-correlates-with-viewership-of-the-big-bang-theory

End.